Experiment 6

Aim : Apply K-Means and Hierarchical Clustering on sample datasets

Dataset : clustering_experiment_data.csv

Theory :

Clustering Analysis

Clustering is a type of **Unsupervised Learning**. Unlike the classification tasks we did earlier (Titanic/Adult Income), there are no "correct answers" or labels provided to the machine. The goal is to let the algorithm find hidden patterns and group similar data points together naturally.

1. K-Means Clustering (The "Centroid" Approach)

K-Means is like a political election where "candidates" (called **Centroids**) try to win over the nearest "voters" (data points).

- **How it works:** You tell the model how many groups you want (K). It places K random points in the data. Every data point joins the group of the closest center point. Then, the center point moves to the "middle" of its new group. This repeats until the centers stop moving.
- **The Goal:** To make sure every point is as close as possible to the center of its cluster.
- **Best for:** Large datasets with well-defined, circular-shaped groups.

2. Hierarchical Clustering (The "Family Tree" Approach)

Instead of picking a number of groups at the start, Hierarchical Clustering builds a multi-level structure of relationships. The version used in this experiment is **Agglomerative (Bottom-Up)**.

- **How it works:** It starts by treating every single data point as its own tiny cluster. It then finds the two closest points and merges them into a pair. It keeps merging the "closest" clusters until everyone is part of one giant group.
- **Linkage (Ward's Method):** In your code, we use the "Ward" method. This is a rule for merging that tries to keep the clusters as tight and compact as possible, similar to K-Means.
- **Best for:** Understanding the relationship between groups or when you aren't sure how many clusters exist.

3. The Dendrogram (The "Map of Merges")

One of the most powerful tools in Hierarchical Clustering is the **Dendrogram**.

- It looks like an upside-down tree or a bracket.
- The **vertical lines** represent the distance between clusters. The longer the line, the more "different" the clusters are.
- By looking at this chart, you can decide where to "cut" the tree to get the number of clusters that makes the most sense for your data.

Code and Outputs :

Step 1 : Implement K-Means Clustering

```
from sklearn.cluster import KMeans

# Initialize KMeans with 4 clusters and random_state 42
kmeans = KMeans(n_clusters=4, random_state=42, n_init='auto')

# Fit the model and predict cluster assignments
df['KMeans_Labels'] = kmeans.fit_predict(df[['Feature 1', 'Feature 2']])

# Display the first few rows to verify labels
print("Dataset with K-Means labels:")
print(df.head())
```

Output :

```
Dataset with K-Means labels:
   Feature 1  Feature 2  KMeans_Labels
0  -9.297689   6.473679              3
1  -9.698741   6.938967              3
2  -1.686653   7.793442              0
3  -7.097308  -5.781333              1
4 -10.876452   6.315437              3
```

Step 2 : Implement Hierarchical Clustering

```
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt
```

SubStep 1. Create a linkage matrix using the 'ward' method

```
Z = linkage(df[['Feature 1', 'Feature 2']], method='ward')
```

SubStep 2. Plot the dendrogram

```
plt.figure(figsize=(10, 7))
plt.title('Hierarchical Clustering Dendrogram')
plt.xlabel('Sample Index')
plt.ylabel('Distance (Ward)')
dendrogram(Z, leaf_rotation=90., leaf_font_size=8.)
plt.show()
```

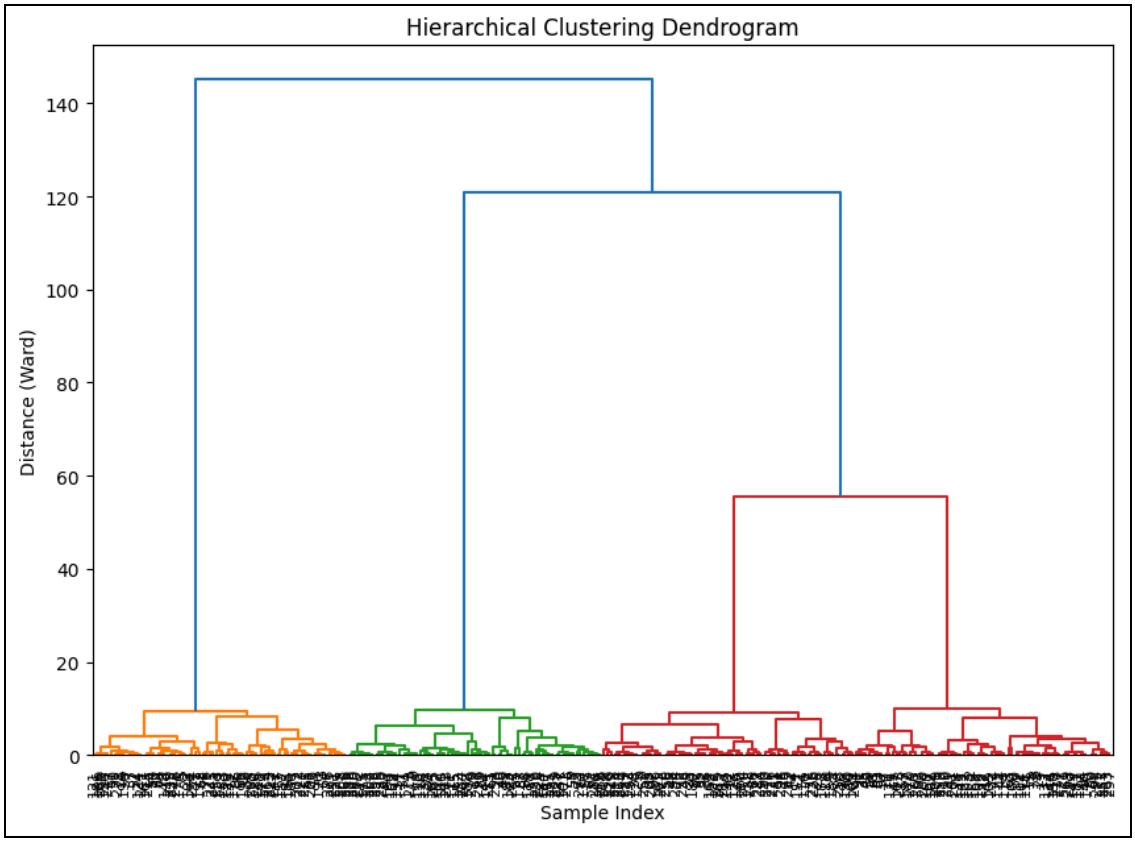
SubStep 3. Instantiate and fit AgglomerativeClustering

```
agg_clustering = AgglomerativeClustering(n_clusters=4)
df['Hierarchical_Labels'] = agg_clustering.fit_predict(df[['Feature 1', 'Feature 2']])
```

SubStep 4. Display the first few rows

```
print("Dataset with Hierarchical Clustering labels:")
print(df.head())
```

Output :




```
Dataset with Hierarchical Clustering labels:
   Feature 1  Feature 2  KMeans_Labels  Hierarchical_Labels
0  -9.297689   6.473679              3                    3
1  -9.698741   6.938967              3                    3
2  -1.686653   7.793442              0                    0
3  -7.097308  -5.781333              1                    1
4 -10.876452   6.315437              3                    3
```

Step 3 : Visualize and Compare the results

```
import matplotlib.pyplot as plt

# Create a figure with two side-by-side subplots
fig, axes = plt.subplots(1, 2, figsize=(15, 6))
```

SubStep 1. Plot K-Means Results

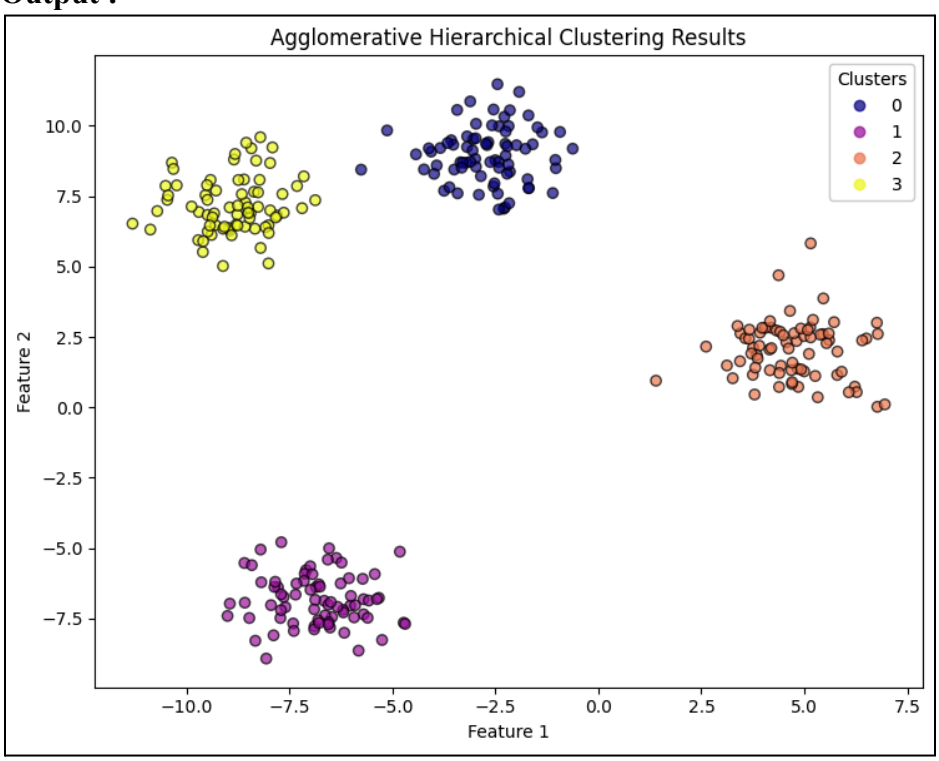
```
scatter1 = axes[0].scatter(df['Feature 1'], df['Feature 2'], c=df['KMeans_Labels'], cmap='viridis',
edgecolors='k', alpha=0.7)
axes[0].set_title('K-Means Clustering Results')
axes[0].set_xlabel('Feature 1')
axes[0].set_ylabel('Feature 2')
legend1 = axes[0].legend(*scatter1.legend_elements(), title="Clusters")
axes[0].add_artist(legend1)
```

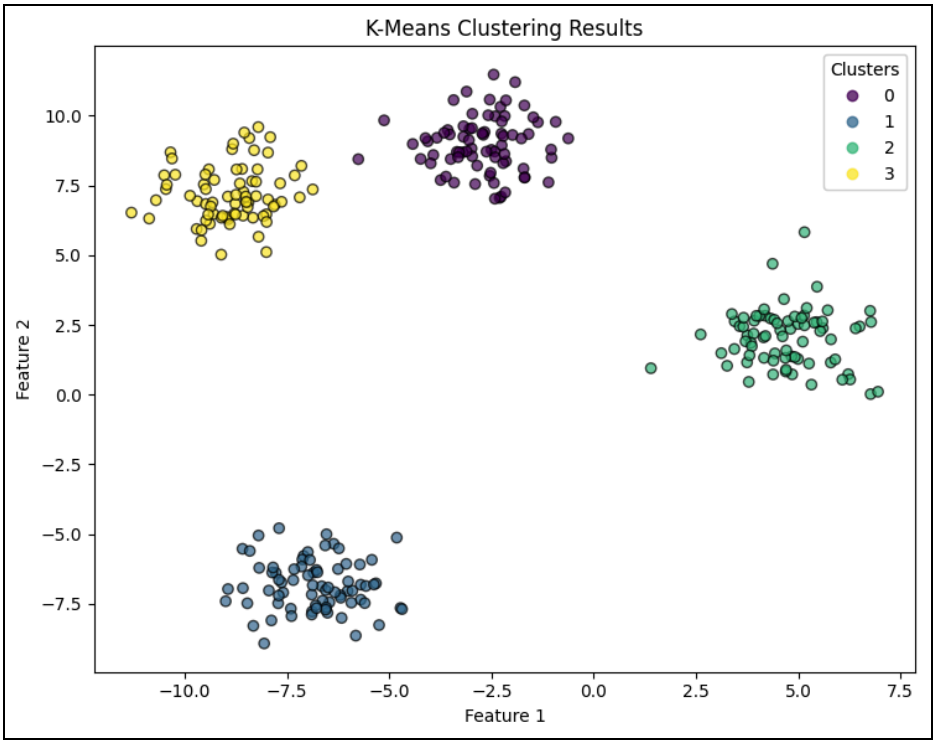
Sub Step 2. Plot Hierarchical Clustering Results

```
scatter2 = axes[1].scatter(df['Feature 1'], df['Feature 2'], c=df['Hierarchical_Labels'],
cmap='plasma', edgecolors='k', alpha=0.7)
axes[1].set_title('Agglomerative Hierarchical Clustering Results')
axes[1].set_xlabel('Feature 1')
axes[1].set_ylabel('Feature 2')
legend2 = axes[1].legend(*scatter2.legend_elements(), title="Clusters")
axes[1].add_artist(legend2)

# Adjust layout and display
plt.tight_layout()
plt.show()
```

Output :





Conclusion : Both methods successfully identified the four distinct clusters generated in the synthetic dataset. While K-Means relies on centroid-based partitioning, the Agglomerative approach uses a bottom-up hierarchical strategy (specifically the Ward linkage method). On this well-separated dataset, both algorithms produced nearly identical cluster boundaries.